PROJECT REPORT

Crunchbase Startup Fundraising Analysis

High-Performance Data Engineering & Memory Optimization Pipeline

Project Overview

This repository contains a production-grade data engineering pipeline designed to process, clean, and analyze high-cardinality historical startup investment data from Crunchbase (October 2013 archive).

When managing large-scale data workflows under strict infrastructure constraints, loading massive flat files directly into memory causes severe bottlenecks. This project solves that problem by implementing an automated **chunk-by-chunk execution strategy**, dynamic type optimization, and seamless relational database migration using **Python, Pandas, and SQLite**.

Technical Architecture & Pipeline Stages

1. Memory Profile Analysis

- Implemented an incremental processing loop using `pandas.read_csv()` restricted to strict **5,000-row chunks** to keep the real-time memory footprint under a critical **1 MB threshold**.
- Conducted comprehensive structural auditing to profile missing data values and track object-type overhead across high-cardinality fields.

2. Data Cleaning & Type Optimization

- Isolated and dropped uninformative high-overhead text fields (`company_permalink`, `investor_permalink`, etc.) to minimize unnecessary relational storage.
- Handled character encoding mismatches dynamically using `latin1` parsing filters.
- Optimized numeric data structures by downcasting integer fields to memory-efficient Pandas formats (`Int32`) and standardizing missing analytical labels to mitigate float64 bloat.

3. Relational Database Schema Design

- Built an analytical data warehouse structure locally inside a standalone **SQLite** database.
- Defined an explicit database schema with designated field configurations.

- Applied targeted **Secondary B-Tree Indexes** (`idx_raised_amount`, `idx_company_category`, `idx_investor_name`) to boost multi-table filters and deliver sub-millisecond query execution speeds.

Analytical Insights & Business Intelligence

Once the relational layers were fully optimized, custom SQL queries were executed via Python to extract market-critical operational insights:

- **Funding Inequality**: Capital distribution follows a heavy Pareto curve. The top **1%** and **10%** of institutional investment rounds secure the vast majority of total market funds, while the bottom **10%** account for a negligible fraction of global startup capital.
- **Sector Dominance**: High-tech sectors, biotechnology, and software engineering capture the largest volume of financing rounds.
- **Institutional Leaders**: Identified the top institutional venture funds driving cumulative capital allocations globally.

Tech Stack & Environment

- **Language:** Python
- **Libraries:** Pandas, NumPy
- **Database Engine:** SQLite (via `sqlite3`)
- **Environment:** Google Colab / Jupyter Notebooks

Acknowledgments

- Historical data sets sourced public archives: datahoarder/crunchbase-october-2013.
- Architectural practices inspired by production-ready enterprise data pipelines.